# Justin Yang Chae

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## RESEARCH INTERESTS

Reinforcement learning for LLM and VLM reasoning: self-play, RL with verifiable rewards (RLVR), gym-style training environments, and long-horizon agents for training-data curation.

## EDUCATION

**University of Washington, Seattle** Sept 2023 – June 2027 (expected)

*B.S. Computer Science and B.A. Mathematics GPA: 3.87/4.0*

- **ML / AI:** Machine Learning (CSE 446), Deep Learning (CSE 493G), Advanced Machine Learning (CSE 493S), Computer Vision (CSE 455), Natural Language Processing (CSE 447), Data Visualization (CSE 442), Machine Learning Capstone (CSE 481M)
- **CS core:** Algorithms (CSE 421), Data Structures & Parallelism (CSE 332), Systems Programming (CSE 333), Hardware/Software Interface (CSE 351), Data Management (CSE 344)
- **Mathematics:** Advanced Linear Algebra (MATH 318), Advanced Multivariable Calculus (MATH 224), Linear Optimization (MATH 407), Discrete Mathematical Modeling (MATH 381), Differential Equations (MATH 207), Matrix Algebra (MATH 208)

**Prior coursework:** UC Berkeley Pre-College Scholars (2021, 4.0) • Bellevue College, Calculus III (2023, 4.0)

## PUBLICATIONS

[1] MD Tanvirul Alam, Saksham Aggarwal, **Justin Yang Chae**, Nidhi Rastogi. "SPHINX: A Synthetic Environment for Visual Perception and Reasoning." *CVPR 2026 Findings Workshop*; earlier version at *Multimodal Algorithmic Reasoning (MAR) Workshop, NeurIPS 2025*. [paper]

[2] **Justin Yang Chae**, MD Tanvirul Alam, Nidhi Rastogi. "Towards Understanding Self-play for LLM Reasoning." *5th Workshop on Mathematical Reasoning and AI (MATH-AI), NeurIPS 2025*. [paper]

## RESEARCH EXPERIENCE

**Prime Intellect** June 2026 – Present

*RL Resident (Peer: Florian Brand)*

- Built PretrainGym, an environment that tests one piece of recursive self-improvement (RSI): a coding agent curates a pretraining corpus for a fixed nanoGPT-style model under a 400M-token budget, rewarded by the trained model's held-out loss minus a decontamination-based leakage penalty. [blog]
- Hardened the environment against reward hacking over three versions after agents inferred the held-out set, first through web search and then through self-scoring; the final version scores on Paloma's 12 sources from a network-free sandbox with an MCP server of five scoped tools.
- Evaluated three frontier coding agents (Kimi K3, GLM-5.3, GPT-6.1 Sol); Kimi K3 scored best (reward 1.042, test loss 3.125) by screening corpora cheaply and tuning mixtures at full budget. Published every run in a turn-by-turn trace viewer.

**AI4SEC Lab, Rochester Institute of Technology** May 2025 – Aug 2025

*REU Intern (Advisors: Prof. Nidhi Rastogi, MD Tanvirul Alam)*

- Led a first-author study of asymmetric proposer–solver self-play post-training on 3B and 7B LLMs with verl, comparing it against RLVR and SFT through parameter-update sparsity, token-entropy dynamics, alternative proposer rewards and pass@k [2].
- Showed the performance ceiling of purely on-policy self-play and the case for off-policy support.
- Co-developed SPHINX, a procedurally generated visual perception gym with 25 task types; frontier VLMs such as GPT-4o score 51.1%, and RLVR training on it improves external visual-reasoning benchmarks [1].

**Ubiquitous Computing Lab, University of Washington** Jan 2025 – Present

*Student Researcher (Advisors: Prof. Shwetak Patel, Vidya Srinivas)*

- Developed multimodal speaker identification from smart-glasses audio and video streams using audio and image encoders.
- Built an egocentric audio memory system that retrieves conversational snippets for personalized chatbots.
- Engineered an ASR and speaker-diarization pipeline (NVIDIA Parakeet, Pyannote) feeding structured transcripts to LLMs for keyword extraction.

- Prototyped a computer-architecture-inspired memory hierarchy for LLMs, with memory tiers at different lookup latencies; benchmarked related memory modules using Ollama, Docker and LangChain.

## HONORS & AWARDS

- Leo Maddox Foundation Endowed Scholarship 2024
- National Merit Scholarship 2023 – 2027
- UW Dean's List, every graded quarter Autumn 2023 – Spring 2026; Annual Dean's List 2023–24, 2024–25

## SERVICE

- Reviewer, MATH-AI: The 6th Workshop on Mathematical Reasoning and AI, NeurIPS 2026 2026
- Reviewer, MATH-AI: The 5th Workshop on Mathematical Reasoning and AI, NeurIPS 2025 2025

## SELECTED PROJECTS

**Merge–Split Dynamic Tokenization** Spring 2026
*CSE 481M Machine Learning Capstone, with Pang Wei Koh and Zhiyuan Zeng*

- Designed inference-time merge–split tokenization; showed entropy-guided splitting recovers accuracy after aggressive token merging in English and Korean.

**Recursive Computation in Continuous Thought Machines** Sept 2025 – Dec 2025
*CSE 455 Computer Vision*

- Trained ablated Continuous Thought Machines on CIFAR-10, improving about 2% over the original paper's baseline and showing not every component is needed for every task.
- Argued that recursive models need more experiments and theory before they become a new scaling axis.

**LLM Post-training for Product Recommendation (Rec-R1 Reproduction)** Mar 2025 – June 2025
*CSE 493S Advanced Machine Learning*

- Reproduced Rec-R1, which trains LLMs with GRPO on a retrieval-match reward; confirmed recommendation gains while general reasoning (e.g., MMLU) held steady.

**Species Distribution Modeling with CNNs** Sept 2024 – Dec 2024
*CSE 493G Deep Learning*

- Built 2D and 3D CNNs on Southern Ocean phytoplankton presence data, reaching 80% (2D) and 79% (3D) accuracy, on par with prior ML methods.

## SKILLS

**Languages:** Python, Java, C, JavaScript/TypeScript, SQL, HTML/CSS, LaTeX

**ML & research tools:** PyTorch, verl, NumPy, Pandas, scikit-learn, Weights & Biases, Matplotlib, Seaborn

**Infrastructure:** Slurm, Apptainer, Docker, Git, Linux, Cloudflare Tunnels

## ACTIVITIES

- Violinist, University of Washington Symphony Orchestra 2023 – Present
- Violinist, Seattle Youth Symphony Orchestra (1st violin section) 2014 – 2023